

CO3-AT1-ANALYTICAL PROBLEM SOLVING

Course: C++ Object Oriented Programming

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Q1. Banking Account Management System

```
#include <iostream>

using namespace std;

class Account
{
    int accountNumber;
    string customerName;
    double balance;
public:
    Account()
    {
        accountNumber = 0;
        customerName = "Unknown";
        balance = 0.0;
        cout << "Default Constructor Called" << endl;
    }
    Account(int accNo, string name, double bal)
    {
        accountNumber = accNo;
        customerName = name;
        balance = bal;
        cout << "Parameterized Constructor Called" << endl;
    }
    Account(const Account &a)
    {
        accountNumber = a.accountNumber;
        customerName = a.customerName;
```

```

        balance = a.balance;

        cout << "Copy Constructor Called" << endl;
    }

    void display()
    {
        cout << "Account Number: " << accountNumber << endl;
        cout << "Customer Name: " << customerName << endl;
        cout << "Balance: " << balance << endl;
        cout << endl;
    }
};

int main()
{
    Account a1;
    a1.display();

    Account a2(101, "Meghana", 50000);
    a2.display();

    Account a3(a2);
    a3.display();

    return 0;
}

```

The screenshot shows a C++ IDE with two panes. The left pane, titled 'main.cpp', contains the source code for an 'Account' class and its 'main' function. The right pane, titled 'Output', shows the results of the program's execution.

Source Code (main.cpp):

```

10 public:
31 {
32     customerName = a.customerName;
34     balance = a.balance;
35     cout << "Copy Constructor Called" << endl;
36 }
37
38 void display()
39 {
40     cout << "Account Number: " << accountNumber << endl;
41     cout << "Customer Name: " << customerName << endl;
42     cout << "Balance: " << balance << endl;
43     cout << endl;
44 }
45 };
46
47 int main()
48 {
49     Account a1;
50     a1.display();
51
52     Account a2(101, "Meghana", 50000);
53     a2.display();

```

Output:

```

Default Constructor Called
Account Number: 0
Customer Name: Unknown
Balance: 0

Parameterized Constructor Called
Account Number: 101
Customer Name: Meghana
Balance: 50000

Copy Constructor Called
Account Number: 101
Customer Name: Meghana
Balance: 50000

=== Code Execution Successful ===

```

Q2. Inventory Product Management System

```
#include <iostream>

using namespace std;

class Product
{
    int id;

    string name;

    int stock;

public:
    Product(int i, string n, int s)
    {
        id = i;

        name = n;

        stock = s;

        cout << "Parameterized Constructor Called for " << name << endl;
    }

    Product(const Product &p)
    {
        id = p.id;

        name = p.name;

        stock = p.stock;

        cout << "Copy Constructor Called for " << name << endl;
    }

    Product operator+(const Product &p)
    {
        Product temp(id, name + " Combined", stock + p.stock);

        return temp;
    }

    ~Product()
    {
        cout << "Destructor Called for " << name << endl;
    }
}
```

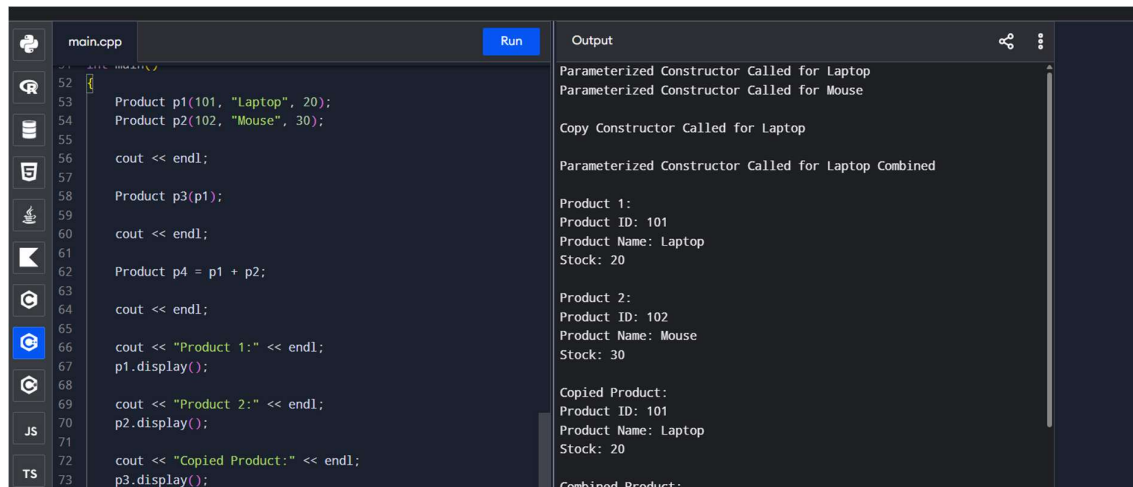
```

void display()
{
    cout << "Product ID: " << id << endl;
    cout << "Product Name: " << name << endl;
    cout << "Stock: " << stock << endl;
    cout << endl;
}

};

int main()
{
    Product p1(101, "Laptop", 20);
    Product p2(102, "Mouse", 30);
    cout << endl;
    Product p3(p1);
    cout << endl;
    Product p4 = p1 + p2;
    cout << endl;
    cout << "Product 1:" << endl;
    p1.display();
    cout << "Product 2:" << endl;
    p2.display();
    cout << "Copied Product:" << endl;
    p3.display();
    cout << "Combined Product:" << endl;
    p4.display();
    return 0;
}

```



```
main.cpp
52 {
53     Product p1(101, "Laptop", 20);
54     Product p2(102, "Mouse", 30);
55
56     cout << endl;
57
58     Product p3(p1);
59
60     cout << endl;
61
62     Product p4 = p1 + p2;
63
64     cout << endl;
65
66     cout << "Product 1:" << endl;
67     p1.display();
68
69     cout << "Product 2:" << endl;
70     p2.display();
71
72     cout << "Copied Product:" << endl;
73     p3.display();
74 }
```

```
Output
Parameterized Constructor Called for Laptop
Parameterized Constructor Called for Mouse

Copy Constructor Called for Laptop

Parameterized Constructor Called for Laptop Combined

Product 1:
Product ID: 101
Product Name: Laptop
Stock: 20

Product 2:
Product ID: 102
Product Name: Mouse
Stock: 30

Copied Product:
Product ID: 101
Product Name: Laptop
Stock: 20

Combined Product:
```

Q3. Shopping Cart Management System

```
#include <iostream>
```

```
using namespace std;
```

```
class ShoppingCart
```

```
{
```

```
    string customerName;
```

```
    int items;
```

```
    double totalAmount;
```

```
public:
```

```
    ShoppingCart(string name, int i, double amount)
```

```
{
```

```
        customerName = name;
```

```
        items = i;
```

```
        totalAmount = amount;
```

```
        cout << "Parameterized Constructor Called" << endl;
```

```
}
```

```
    ShoppingCart(const ShoppingCart &c)
```

```
{
```

```
        customerName = c.customerName;
```

```
        items = c.items;
```

```
        totalAmount = c.totalAmount;
```

```
        cout << "Copy Constructor Called" << endl;
```

```

    }

ShoppingCart operator+(const ShoppingCart &c)
{
    ShoppingCart temp(
        customerName + " " + c.customerName,
        items + c.items,
        totalAmount + c.totalAmount
    );
    return temp;
}

~ShoppingCart()
{
    cout << "Destructor Called for " << customerName << endl;
}

void display()
{
    cout << "Customer: " << customerName << endl;
    cout << "Number of Items: " << items << endl;
    cout << "Total Amount: " << totalAmount << endl;
    cout << endl;
}

};

int main()
{
    ShoppingCart cart1("Meghana", 3, 1500);
    ShoppingCart cart2("Rahul", 2, 1000);
    cout << endl;
    ShoppingCart cart3(cart1);
    cout << endl;
    ShoppingCart cart4 = cart1 + cart2;
    cout << endl;
}

```

```

    cout << "Cart 1:" << endl;

    cart1.display();

    cout << "Cart 2:" << endl;

    cart2.display();

    cout << "Copied Cart:" << endl;

    cart3.display();

    cout << "Merged Cart:" << endl;

    cart4.display();

    return 0;
}

```

The screenshot shows a C++ IDE with a file named `main.cpp`. The code in the editor is as follows:

```

57 {
58     ShoppingCart cart1("Meghana", 3, 1500);
59     ShoppingCart cart2("Rahul", 2, 1000);
60
61     cout << endl;
62
63     ShoppingCart cart3(cart1);
64
65     cout << endl;
66
67     ShoppingCart cart4 = cart1 + cart2;
68
69     cout << endl;
70
71     cout << "Cart 1:" << endl;
72     cart1.display();
73
74     cout << "Cart 2:" << endl;
75     cart2.display();
76
77     cout << "Copied Cart:" << endl;
78     cart3.display();

```

The output window on the right shows the following results:

```

Customer: Meghana
Number of Items: 3
Total Amount: 1500

Cart 2:
Customer: Rahul
Number of Items: 2
Total Amount: 1000

Copied Cart:
Customer: Meghana
Number of Items: 3
Total Amount: 1500

Merged Cart:
Customer: Meghana + Rahul
Number of Items: 5
Total Amount: 2500

Destructor Called for Meghana + Rahul
Destructor Called for Meghana
Destructor Called for Rahul
Destructor Called for Meghana

```